

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

LABORATORY CORPORATION OF AMERICA HOLDINGS,
Petitioner,

v.

RAVGEN, INC.,
Patent Owner.

IPR2021-01026
Patent 7,727,720 B2

Before ZHENYU YANG, TIMOTHY G. MAJORS, and DAVID COTTA,
Administrative Patent Judges.

COTTA, *Administrative Patent Judge.*

JUDGMENT

Final Written Decision

Determining No Challenged Claims Unpatentable

35 U.S.C. § 318(a)

Granting Patent Owner's Motion for Entry of Protective Order and to Seal;

Granting Petitioner's Motion to Seal

37 C.F.R. §§ 42.14, 42.54

I. INTRODUCTION

On May 28, 2021, Laboratory Corporation of America Holdings (“Petitioner” or “Labcorp”),¹ filed a Petition to institute *inter partes* review of claims 1–9 and 21–30 of U.S. Patent No. 7,727,720 B2 (Ex. 1001, “the ’720 patent”). Paper 1 (“Pet.” or “Petition”). We instituted trial on December 14, 2021. Paper 11 (“Inst. Dec.”). During trial, Ravgen, Inc. (“Patent Owner”)² filed a Patent Owner Response. Paper 20 (“PO Resp.”).³ Later filings include Petitioner’s Reply (Paper 30 (“Reply Br.”))⁴ and Patent Owner’s Sur-Reply (Paper 32 (“Sur-reply”)). An oral hearing was held on September 21, 2022, and a transcript is entered in the record. Paper 50 (“Tr.”).

We have jurisdiction under 35 U.S.C. § 6(b). After considering the parties’ arguments and evidence, we determine that Petitioner has not proved by a preponderance of the evidence that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e). Our reasoning is explained below, and we issue this Final Written Decision under 35 U.S.C. § 318(a).

¹ Petitioner identifies itself, Resolution Bioscience, and Agilent Technologies as the real parties-in-interest. Pet. 1.

² Patent Owner identifies itself as the real party-in-interest. Paper 4, 1.

³ Paper 20 is the redacted public version of Patent Owner’s Response. The non-public version (Paper 19) was filed together with Patent Owner’s Motion for Entry of the Stipulated Protective Order and to Seal Patent Owner’s Response and Exhibits 2080, and 2170–73 (Paper 21).

⁴ Paper 30 is the redacted public version of Petitioner’s Reply. The non-public version (Paper 29) was filed together with Petitioner’s Motion to Seal Petitioner’s Reply (Paper 28).

A. Related Patents & Proceedings

The '720 patent issued on June 1, 2010, from U.S. Patent Application No. 11/212,812 (“the '812 Application”), filed August 26, 2005. Ex. 1001, codes (21), (22), (45). The '812 Application is a continuation of Application No. PCT/US2004/006337, filed March 1, 2004, which itself claims priority through a chain of several continuation and continuation-in-part applications with filing dates in 2003, and a provisional application filed May 8, 2002. *Id.* at codes (60), (63). Related U.S. Patent No. 7,332,277 (“the '277 patent”) issued on February 19, 2008, and claims priority to common applications. *Id.*; IPR2021-00788, Exhibit 1001 at codes (60), (63).

The parties identify multiple lawsuits involving the '277 and '720 patents. Pet. 1–2; Paper 4, 1. Prior to institution, the parties both identified *Ravgen, Inc. v. Laboratory Corp. of America Holdings*, No. 6:20-cv-00969-ADA (W.D. Tex.) (“the Labcorp lawsuit”), as relating to the present proceeding. *Id.* After we instituted review, Petitioner and Patent Owner stipulated to dismiss Count II of Patent Owner’s complaint in the Labcorp litigation with prejudice. Paper 45. Prior to dismissal, Patent Owner had asserted that Labcorp infringed the '720 patent. Pet. 1–2; Paper 4, 1. With the dismissal, “the '720 patent is no longer at issue in the litigation between the parties.” Paper 44 (Updated Mandatory Notice, citing Ex. 1077).

According to the parties, Patent Owner has also asserted the '720 patent against other defendants such as Natera, Inc. and NSTX, Inc., Quest Diagnostics Inc., PerkinElmer, Inc., Myriad Genetics, Inc., Progenity, Inc., Illumina Inc., and Ariosa Diagnostics, Inc. in other lawsuits pending in various district courts around the country. Pet. 1–2; Paper 4, 1. The parties

also identify other matters involving the '720 patent pending before the Patent Office. *Id.* The '720 patent was challenged in IPR2021-00791 (filed by Quest Diagnostics), and IPR2021-01271 (filed by Illumina, Inc.).⁵ Pet. 2; Paper 4, 1; Paper 6, 1. In addition to the above, the parties identify Ex Parte Reexamination Control Nos. 90/014,703 and 90/014,869 as relating to the '720 patent. Pet. 2; Paper 4, 1; Paper 13, 1. Both reexaminations have been stayed. Paper 17; IPR2021-00791, Paper 25.

B. Asserted Grounds of Unpatentability

Petitioner asserts four grounds of unpatentability in this Petition (Pet. 9), which are provided in the table below:

Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1–6, 21, 23	102(b) ⁶	Lee ⁷
22	103(a)	Lee
7–9, 24–30	103(a)	Lee, Sorenson ⁸

⁵ The Board terminated IPR2021-00791 on October 12, 2022, based on a settlement between the parties in that proceeding. IPR2021-01271 remains pending.

⁶ The Leahy-Smith America Invents Act, Pub. L. No. 112-29, 125 Stat. 284 (2011) (“AIA”), amended 35 U.S.C. §§ 102 and 103. Based on the filing date of the '720 patent, we apply the pre-AIA versions of §§ 102 and 103.

⁷ Lee et al., *Quantitation of Genomic DNA in Plasma and Serum Samples: Higher Concentrations of Genomic DNA Found in Serum than in Plasma*, 41 *Transfusion*, 276–282 (2001) (Ex. 1004, “Lee”).

⁸ Sorenson et al., *Soluble Normal and Mutated DNA Sequences from Single-Copy Genes in Human Blood*, 3 *Cancer Epidemiology, Biomarkers & Prevention*, 67–71 (1994) (Ex. 1005, “Sorenson”).

Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
1–9, 21–30	103(a)	Sorenson, Rao ⁹

Petitioner relies on two declarations of Dr. Jeremy S. Edwards, among other evidence. Ex. 1002; Ex. 1059. Patent Owner responds with a declaration from Dr. Brian Van Ness, along with other evidence. Ex. 2078.¹⁰

C. Technology Overview & the '720 Patent

The '720 patent discloses a “rapid, non-invasive method for determining the sequence of DNA from a fetus” that can be used “for detection of chromosomal abnormalities.” Ex. 1001, 1:23–35. The '720 patent explains that a variety of invasive and non-invasive techniques are available for prenatal diagnosis, including amniocentesis, fetal blood cells in maternal blood, and maternal serum alpha-fetoprotein. *Id.* at 2:42–46. According to the patent, however, “techniques that are non-invasive are less specific, and the techniques with high specificity and high sensitivity are highly invasive.” *Id.* at 2:46–49; *see also id.* at 3:24–27 (citing an increased fetal mortality risk of about 0.5% with amniocentesis).

The '720 patent acknowledges the prior non-invasive use of fetal cells and cell-free fetal DNA (“cffDNA”), both isolated from maternal blood, for

⁹ Rao et al., PCT Publication No. WO 03/018757, published March 6, 2003 (Ex. 1006, “Rao”).

¹⁰ Patent Owner submitted the Declaration of Dr. Glenn D. Prestwich (Ex. 2001, “the Prestwich Declaration”) in connection with its Preliminary Response.

prenatal diagnosis. *Id.* at 4:59–5:44. With regard to fetal cells, the patent notes that the “presence of fetal nucleated cells in maternal blood makes it possible to use these cells for noninvasive prenatal diagnosis,” and that such “cells can be sorted and analyzed by a variety of techniques to look for particular DNA sequences.” *Id.* at 4:60–65. Yet the patent states that “it is still difficult” to get many fetal cells from maternal blood and “[t]here may not be enough to reliably determine anomalies of the fetal karyotype or assay for other abnormalities.” *Id.* at 5:15–21. The ’720 patent states that fetal DNA “has been detected and quantitated in maternal plasma and serum” and that “fetal DNA present in the maternal serum and plasma is comparable to the concentration of DNA obtained from fetal cell isolation protocols.” *Id.* at 5:24–34. “However,” according to the patent, “the diagnostic and clinical applications of circulating fetal DNA is limited to genes that are present in the fetus but not the mother” and that “a need still exists for a non-invasive method that can determine the sequence of fetal DNA and provide definitive diagnosis of chromosomal abnormalities in a fetus.” *Id.* at 5:38–44.

The ’720 patent describes a method that is said to increase the proportion or percentage of the cffDNA component in a sample from a pregnant female for subsequent analysis. According to the ’720 patent, the ability to detect chromosomal abnormalities has been “hindered by the low percentage of free fetal DNA” in maternal samples. Ex. 1001, 89:26–30. “Increasing the percentage of free fetal DNA would enhance the detection” of mutations, trisomy, and other genetic abnormalities. *Id.* at 89:30–35.

With the aim of increasing the percentage of cffDNA relative to

circulating maternal DNA in a maternal sample, the '720 patent describes adding an agent that inhibits cell lysis. *Id.* at 210:19–24 (Example 15) (“[T]he use of cell lysis inhibitors, cell membrane stabilizers, or cross-linking reagents can be used to increase the percentage of fetal DNA in the maternal blood.”); *id.* at 89:35–37 (“The percent of fetal DNA in plasma obtained from a pregnant female was determined both in the absence and presence of inhibitors of cell lysis”). The '720 patent explains that, “[w]hile lysis of both maternal and fetal cells is inhibited, the vast majority of cells [in a maternal blood sample] are maternal, and thus by reducing the lysis of maternal cells, there is a relative increase in the percentage of free fetal DNA.” *Id.* at 33:42–45. The patent identifies numerous agents as cell lysis inhibitors, cell membrane stabilizers, or cross-linking reagents. *See, e.g., id.* at 32:65–33:28 (listing, for example, formaldehyde, formalin, cleavable crosslinkers, cholesterol, vitamin E and its derivatives, and glucose); *see also id.* at 6:49–65; 11:43–12:5; 216:14–218:55 (Table XXIII (“A representative list of cross-linkers that can be used to impede maternal cell lysis”)).

The '720 patent provides results on the use of formalin (i.e., formaldehyde in aqueous solution) as the lysis-inhibiting agent. *Id.* at 89:25–92:26 (Example 4), 210:16–218:55 (Example 15). In Example 4, the patent describes collecting a 5 ml blood sample from a pregnant subject, separating the sample into two tubes (each containing EDTA¹¹), and adding

¹¹ The acronym “EDTA” refers to ethylenediaminetetracetic acid. *See* Ex. 1060, Abstract. The '720 patent states that EDTA is a “magnesium chelator.” Ex. 1001, 32:61–64.

formaldehyde (25µl/ml) to one of the tubes. *Id.* at 89:40–48. The samples were centrifuged and 800 µl of each maternal plasma sample was then used for DNA purification and further processing to determine the relative amount of cffDNA in the samples. *Id.* at 89:47–92:26. According to the ’720 patent, “the percentage of fetal DNA present in the sample that was treated with only EDTA was 1.56%” and the “percentage of fetal DNA present in the sample treated with formalin and EDTA was 25%.” *Id.* at 91:34–41. The patent reports that the percent of total cffDNA in eighteen blood samples with and without formalin was then calculated, with the results (mean percentage cffDNA) provided in Table V. *Id.* at 91:45–92:10 (reporting, *inter alia*, a mean percentage fetal DNA of 19.47% with formalin versus 7.71% without formalin); *see also id.* at 210:16–215:65 (Example 15 (reporting further results on formalin use in Table XXI)).

D. Challenged Claims

The ’720 patent includes thirty claims. Of those, claims 1–9 and 21–30 are challenged here. Pet. 9. Claim 1, the only independent claim, is illustrative of the claims challenged in this Petition, and reads as follows:

1. A method for detecting a free nucleic acid, wherein said method comprises: (a) isolating free nucleic acid from a non-cellular fraction of a sample, wherein said sample comprises an agent that impedes cell lysis, if cells are present, and wherein said agent is selected from the group consisting of membrane stabilizer, cross-linker, and cell lysis inhibitor; and (b) detecting the presence or absence of the free nucleic acid.

Ex. 1001, 535:15–21.

II. ANALYSIS

A. *Principles of Law*

“In an [*inter partes* review], the petitioner has the burden from the onset to show with particularity why the patent it challenges is unpatentable.” *Harmonic Inc. v. Avid Tech., Inc.*, 815 F.3d 1356, 1363 (Fed. Cir. 2016) (citing 35 U.S.C. § 312(a)(3) (requiring *inter partes* review petitions to identify “with particularity . . . the evidence that supports the grounds for the challenge to each claim”)).

A claim is unpatentable under 35 U.S.C. § 103(a) if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which that subject matter pertains. *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). The question of obviousness is resolved on the basis of underlying factual determinations including: (1) the scope and content of the prior art; (2) any differences between the claimed subject matter and the prior art; (3) the level of ordinary skill in the art; and (4) secondary considerations of nonobviousness when presented. *Graham v. John Deere Co.*, 383 U.S. 1, 17–18 (1966).

B. *Person of Ordinary Skill in the Art (“POSA”)*

In determining the level of skill in the art, we consider the problems encountered in the art, the art’s solutions to those problems, the rapidity with which innovations are made, the sophistication of the technology, and the

educational level of active workers in the field. *Custom Accessories, Inc. v. Jeffrey-Allan Indus., Inc.*, 807 F.2d 955, 962 (Fed. Cir. 1986).

Petitioner contends a POSA:

would have had an advanced degree (e.g., M.S. or Ph.D.) in Chemistry, Biochemistry, Molecular Biology, Genetics, Bioengineering, Chemical Engineering, or a related discipline, and at least 2-3 years of experience in a research or clinical laboratory. . . . In addition, a [POSA] would have been familiar with the available techniques for optimizing biological samples to be used in various laboratory analyses, such as for detection of cell-free nucleic acids, and would have been familiar with the relevant scientific field and its literature at the time when the application was filed, in particular, literature regarding the detection of cell-free nucleic acids.

Pet. 22 (citing Ex. 1002 ¶¶ 12–13). Patent Owner does not challenge Petitioner’s definition of the POSA, but Patent Owner’s expert, Dr. Van Ness, testifies that a POSA would have had “a M.D. and/or a Ph.D. in a related area such as genetics, biochemistry, molecular biology, cell biology, or microbiology and at least one to two years of work in one of those related areas.” Ex. 2078 ¶ 23; PO Resp. 7. Dr. Van Ness asserts that a POSA could, alternatively, “have a Bachelor’s degree in one of the foregoing areas and at least three to four years of work in” such areas. Ex. 2078 ¶ 23.

Dr. Van Ness’s proposed POSA level is too broad. Under that proposal, an individual with, for example, an undergraduate degree in microbiology and three years’ work experience studying the habitats of bacteria might qualify as a POSA. It is not clear that such a person would have sufficient, relevant experience in the detection of chromosomal abnormalities, especially through non-invasive methods for detecting fetal

genetic abnormalities in maternal samples, as described in the '720 patent. *See, e.g.*, Ex. 1001, 1:22–5:44 (Field of Invention and Background Art). Petitioner's proposal of the POSA level is more precise because it requires familiarity with techniques for detecting cell-free DNA in biological samples, which is relevant to the prior art and the claimed technology. Petitioner's proposal also appears to be more consistent with the cited prior art. *Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) (explaining that specific findings regarding ordinary skill level are not required where the prior art itself reflects an appropriate level and a need for testimony is not shown). Accordingly, we adopt Petitioner's POSA level here, but note that our other determinations on this record would not change under Patent Owner's POSA level.¹²

C. Claim Construction

We interpret a claim “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.100(b) (2020). Under this standard, we construe the claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” *Id.*

The specification may reveal a special definition given to a claim term by the patentee, or the specification or prosecution history may reveal an

¹² Patent Owner's declarant, Dr. Van Ness, testifies that he does not see his definition of the POSA as “materially different” from that offered by Petitioner. Ex. 2078 ¶ 25.

intentional disclaimer or disavowal of claim scope by the inventor. *Phillips v. AWH.*, 415 F.3d 1303, 1316 (Fed. Cir. 2005). If an inventor acts as his or her own lexicographer, the definition must be set forth in the specification with reasonable clarity, deliberateness, and precision. *Renishaw PLC v. Marposs Societa' per Azioni*, 158 F.3d 1243, 1249 (Fed. Cir. 1998). The disavowal, if any, can be effectuated by language in the specification or the prosecution history. *Poly-America, L.P. v. API Indus., Inc.*, 839 F.3d 1131, 1136 (Fed. Cir. 2016). “To disavow claim scope, the specification must contain ‘expressions of manifest exclusion or restriction, representing a clear disavowal of claim scope.’” *Retractable Techs., Inc. v. Becton, Dickinson & Co.*, 653 F.3d 1296, 1306 (Fed. Cir. 2011) (quoting *Epistar Corp. v. Int’l Trade Comm’n*, 566 F.3d 1321, 1335 (Fed. Cir. 2009)). “[T]he standard for disavowal is exacting, requiring clear and unequivocal evidence that the claimed invention includes or does not include a particular feature.” *Poly-America, L.P.*, 839 F.3d at 1136 (citing *Omega Eng’g, Inc. v. Raytek Corp.*, 334 F.3d 1314, 1324 (Fed. Cir. 2003)).

Claim 1 requires the presence of “an agent that impedes cell lysis, if cells are present.” Ex. 1001, 535:17–18. Patent Owner contends that we should construe this limitation to mean: “a substance that inhibits¹³ the rupture of cells that is selected from the group consisting of membrane stabilizer, cross-linker and cell lysis inhibitor, and does not include chelators

¹³ The ’277 patent includes claims that require “an agent that inhibits cell lysis, if cells are present.” IPR2021-00902, Ex. 1001, 473:2–3. The parties use the terms “inhibits” and “impedes” interchangeably.

used as anticoagulants nor endogenous substances.” PO Resp. 7.¹⁴

Petitioner does not offer a claim construction of its own, arguing only that the Board should reject Patent Owner’s proposed construction because it is “irrelevant to the issues in this proceeding.” Reply 2; Tr. 8 (Petitioner’s counsel, responding to question on whether Petitioner was offering “any construction of the agent term,” stating: “No, Your Honor. Its plain and ordinary meaning. We are only interpreting the claim according to plain and ordinary meaning and of course the specification.”).

The question presented by Patent Owner’s claim construction is whether the claimed agent that impedes cell lysis may be a chelator used as an anticoagulant.¹⁵ The specification and prosecution history make clear that one particular anticoagulant chelator – ethylenediaminetetraacetic acid

¹⁴ The district court in *Ravgen v. Natera, Inc. and NSTX, Inc.*, 1:20-cv-00692 (W.D. Tex) adopted this construction. Ex. 2075. We take this claim construction into account into consideration in reaching our own claim construction. 37 C.F.R. § 42.100 (“Any prior claim construction determination concerning a term of the claim in a civil action . . . that is timely made of record in the *inter partes* review proceeding will be considered.”).

¹⁵ Patent Owner proposed construction also excludes “endogenous substances.” PO Resp. 7. We do not consider here whether the agent that impedes cell lysis may include “endogenous substances” because, as Patent Owner concedes, there are “no invalidity grounds in this case that rely on endogenous substances.” Tr. 38; *See Nidec Motor Corp. v. Zhongshan Broad Ocean Motor Co. Ltd.*, 868 F.3d 1013, 1017 (Fed. Cir. 2017) (“[W]e need only construe terms ‘that are in controversy, and only to the extent necessary to resolve the controversy.’” (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999))).

(“EDTA”) – is excluded from the scope of “agent[s] that impedes cell lysis” recited in claim 1. In particular the specification describes an experiment using tubes containing EDTA as a control that demonstrates the “absence ... of inhibitors of cell lysis.” Ex. 1001, 89:35–37, 89:45–47, 91:16-20. And during prosecution of a related application from which the ’720 patent claims priority,¹⁶ Patent Owner overcame a rejection where the Examiner relied on EDTA as the claimed anticoagulant chelator by arguing that “EDTA is not an ‘agent that inhibits cell lysis.’” Ex. 2064, 1196; *see also, id.* at 1024 (Examiner interview summary, reporting that “applicant pointed out [that] EDTA could not be defined as a cell lysis inhibitor but rather was simply an anticoagulant.”). Given this clear record, there is no dispute that EDTA is not an agent that impedes lysis, as recited in claim 1. *See* Tr. 8 (“[Board]: Okay. Do you dispute that EDTA would be excluded from the scope of agent? [Petitioner’s counsel]: No, Your Honor. We agree that EDTA would not be the agent and Petitioner has never argued that EDTA is within the scope of the agent term.”).

Having determined that Patent Owner disavowed EDTA as being the claimed agent that “impedes cell lysis,” we next consider whether this disavowal extends more broadly to other anticoagulants— specifically, as relevant here, acid citrate dextrose (“ACD”). On this record, we find that it does. Importantly, in distinguishing EDTA from the claimed agent, Patent

¹⁶ U.S. Patent Application No. 10/661,165 (“the ’165 application”), issued as the ’277 patent.

Owner relied not on properties unique to EDTA, but on the fact that it was a “well-known chelator” and anticoagulant:

[T]he assertion by the Office that EDTA is a cell lysis inhibitor is simply incorrect. . . EDTA is not an “agent that inhibits cell lysis.” Rather, EDTA is a well-known chelator of calcium and magnesium. EDTA is routinely added to blood during the blood collection process as an anticoagulant due to its ability to chelate calcium.

Ex. 2064, 1196. Given Patent Owner’s clear intent to exclude EDTA from the scope of the claimed “agent” on the basis that it was a chelator and an anticoagulant, we find that a POSA would have understood other chelators and anticoagulants to also be excluded from the scope of the claimed agent.

Consistent with this understanding, during subsequent prosecution of the application that issued as the ’720 patent, Patent Owner argued that the Holodniy reference, which disclosed a method of detecting free nucleic acid in which blood was collected in tubes containing ACD, “*fail[ed] to teach or suggest a method for detecting free nucleic acid comprising, inter alia, isolating free nucleic acid from a non-cellular fraction of a sample, wherein an agent that impedes cell lysis has been added.*” Ex. 1003, 1428 (emphasis added). On the salient point here, the Examiner agreed, concluding that “Holodniy et al. teach a method of detecting free nucleic acid which comprises all of the limitations of Claim 1, *except these authors do not teach that their samples comprise an agent that impedes cell lysis.*” *Id.* at 1344 (emphasis added); *see also id.* at 1156 (finding that Adams et al., a reference that also collects blood in tubes containing ACD, taught “all of the limitations recited in Claim 1 except these authors do not teach adding an agent that impedes cell lysis to the sample.”). And Patent Owner argues

persuasively, without dispute from Petitioner, that ACD is the “**only**” addition to the collection tubes besides the blood sample itself. Sur-reply 4 (citing Ex. 1016, 2; Ex. 2023, 20:52–54). This supports that neither the Examiner nor the Patent Owner considered ACD as an anticoagulant chelator to be an “agent,” as that term is used in claim 1.

Accordingly, we construe the phrase “agent that impedes cell lysis” to exclude chelators used as anticoagulants. Although the agent that impedes lysis cannot itself be an anticoagulant/chelator, the “sample” that comprises “an agent that impedes cell lysis” may still include an anticoagulant/chelator. This is because claim 1 uses the transitional term “comprises,” and is thus open to other components in addition to the claimed agent. Ex. 1001, 535:15–21 (reciting “[a] method for detecting a free nucleic acid, wherein said method *comprises*: (a) isolating free nucleic acid from a non-cellular fraction of a sample, wherein said sample *comprises* an agent that impedes cell lysis”) (emphasis added); *Exergen Corp. v. Wal-Mart Stores, Inc.*, 575 F.3d 1312, 1319 (Fed. Cir. 2009) (“The claim uses the term ‘comprising,’ which is well understood in patent law to mean ‘including but not limited to.’”). Patent Owner does not contend otherwise. See Sur-reply 3 (arguing that “the fact that the **sample** can contain an anticoagulant has nothing to do with how the ‘**agent**’ limitation should be construed.”).

D. Overview of the Asserted Prior Art

1. Lee (Exhibit 1004)

Lee is an article about quantitating cell-free genomic DNA in serum and in plasma. See generally Ex. 1004. Lee states that “a protocol to

process serum and plasma samples for genomic DNA PCR amplification has been optimized, and baseline concentrations of cell-free DNA in serum and plasma have been evaluated for the study of posttransfusion chimerism.” *Id.* at 277. In this study, “[f]resh blood from healthy donors was collected into tubes with ACD (yellow-top), EDTA (purple-top), or no anticoagulant (red-top).” *Id.* The blood was centrifuged and separated into plasma and serum samples. *Id.* After centrifuging, aliquots were prepared from the sample and frozen at -80° C. *Id.* The remaining original sample was stored at 4° C for 24 hours. *Id.* After this period, the original sample was centrifuged again and another aliquot was prepared and frozen at -80° C. *Id.* The remaining original sample again stored at 4° C for 24 hours. This process was repeated for seven days. *Id.* Eventually, the aliquots were thawed, and DNA was extracted and quantified using a PCR assay. *Id.* at 277–278.

Lee discloses detecting cell-free DNA in the samples prepared as described above. *Id.* at 278 (“Paired 2-day-old plasma (ACD) and serum samples collected from 18 normal blood donors were processed and amplified with HLA DQ- α primers (GH26, GH27) for initial evaluation of the concentration of cell-free genomic DNA.”).

2. Sorenson (Exhibit 1005)

Sorenson is an article published in 1994 in the January/February volume of Cancer Epidemiology, Biomarkers and Prevention. Ex. 1005. Sorenson discloses that “[h]ealthy individuals have soluble (extracellular) DNA in their blood, and increased amounts are present in cancer patients.” *Id.* at Abstr. Sorenson reports the “detection of specific sequences of the cystic fibrosis and K-*ras* genes in plasma DNA from normal donors.” *Id.* In

addition, Sorenson identifies “mutated K-*ras* sequences . . . from three patients with pancreatic carcinoma that contain mutated K-*ras* genes.” *Id.* According to Sorenson, “[t]hese results indicate that sequences of single-copy genes can be identified in normal plasma and that the sequences of mutated oncogenes can be detected and identified . . . from patients with malignant tumors containing identical mutated genes.” *Id.* Sorenson teaches that identifying mutated genes in plasma or serum “could be useful for diagnosis, determining response to treatment, and predicting prognosis.” *Id.*

Sorenson discloses that “cellular DNA can be released during clotting of whole blood[,] causing assays of DNA in serum to yield falsely high results.” *Id.* at 67. Sorenson reports that “PCR using primers with 3’-ends complementary to specific genetic point mutations, allows the rapid detection of such mutations in genomic DNA even with a high background of DNA containing normal alleles.” *Id.* at 69.

3. Rao (Exhibit 1006)

Rao is an international patent application that published on March 6, 2003. Ex. 1006, code (43). Rao relates to “[c]ompositions and methods for stabilizing rare cells in blood specimens,” and compositions that “serv[e] as cell fixatives” to “minimize losses of target cells (for example, circulating tumor cells [(CTCs)]) and formation of debris and aggregates from target cells.” *Id.* at Abstr.

Rao discloses that tumor cells undergoing apoptosis have altered membrane permeability that may lead to the escape of DNA and other cellular components. *Id.* at 2:5–10 (“Such tumor cell debris may still bear

epitopes that are characteristic of intact cells, and can lead to spurious increases in circulating cancer cells.”). Moreover, Rao teaches, “[l]eukocytes . . . are known to be labile and diminish on storage,” thus “increas[ing] the amount of cellular debris[] derived from normal blood cells or proteins” and “interfer[ing] with the isolation and detection of rare target cells such as CTC.” *Id.* at 2:14–17. Rao discloses that “[t]here is a large body of published or patented art regarding the stability and stabilization of normal blood cells over time and several proprietary commercial stabilizers are available for preserving white blood cells.” *Id.* at 3:4–8 (listing stabilizers, including “Cyto-Chex™ from Streck Laboratories”). Rao further discloses that the “ideal” stabilizer or preservative preserves the target cells while minimizing formation of interfering cellular debris in a sample. *Id.* at 7:20–24. Rao identifies “the formaldehyde donor imidazolidinyl urea” as an effective stabilizer at a preferred concentration of 0.1–10% by volume of the specimen. *Id.* at 8:2–5; *see also id.* at 22:1–14 (claim 1 (reciting a composition consisting of an anti-coagulating agent and a stabilizing agent), claim 6 (reciting the composition of claim 1 wherein the stabilizing agent “is a formaldehyde donor”)).

E. Anticipation by Lee

Petitioner asserts that claims 1–6, 21, and 23 are unpatentable as anticipated by Lee. Pet. 22–31. We provided an overview of the prior art above. Petitioner contends that each of the limitations of the challenged claims is disclosed in Lee. *Id.* With respect to the limitation of claim 1 requiring an “agent that impedes cell lysis, if cells are present,” Petitioner

points to Lee’s disclosure of blood collection tubes containing ACD. Pet. 25–26. Petitioner argues that the dextrose in ACD meets the “agent” limitation. *Id.* Patent Owner argues that Lee does not disclose an “agent that impedes cell lysis” but does not contest that Lee discloses the other elements recited in claim 1. PO Resp. 15–25. Accordingly, we focus our discussion of the “agent” limitation.

There is no dispute that ACD includes dextrose and that dextrose is a species of glucose. Ex. 1054, 125:3–10 (Dr. Van Ness testimony identifying dextrose as one of the components of ACD and agreeing that dextrose is a form of glucose); Ex. 1036 (disclosing that dextrose is a naturally occurring form of glucose); Ex. 1002 ¶ 58 (Dr. Edwards testimony that “[d]extrose . . . is the dextrorotary form of glucose.”).¹⁷ Nor is there any dispute that the ’720 patent identifies glucose as an agent that impedes cell lysis. *See e.g.*, Ex. 1001, 6:49–65 (“An agent that stabilizes cell membranes may be added to the maternal blood sample to reduce maternal cell lysis including but not limited to . . . glucose”). Rather the dispute here centers on two questions: 1) whether the exclusion of anticoagulant/chelators from qualifying as the claimed agent operates to exclude dextrose introduced to the sample through ACD, and 2) whether Petitioner has established that the dextrose in ACD actually impedes cell lysis in Lee’s method. We address each of these two questions in turn.

¹⁷ Henceforth, when discussing Lee, we use the term “dextrose” and when discussing the ’720 patent we use the term “glucose.” It is our understanding, however, that for purposes of our analysis, the terms are largely interchangeable. *See* Ex. 1002 ¶ 58.

1. Does the exclusion of anticoagulant/chelators from qualifying as the claimed “agent” operate to exclude glucose introduced to a blood sample as part of ACD?

As discussed *infra* § II.C, the claims exclude anticoagulants from being the claimed “agent that impedes cell lysis, if cells are present.” We now consider the application of that exclusion to the individual components of ACD. Before doing so, we provide a brief technical background on ACD.

ACD is a solution “comprised of three components . . . : sodium citrate, citric acid, and dextrose.” Ex. 2001 ¶ 169 (Dr. Prestwich testimony); Ex. 2071 (FDA labeling for ACD). “The sodium citrate component of ACD chelates calcium ions, which prevent them from being used in the clotting process.” Ex. 2001 ¶ 169; Ex. 1054, 125:3–8 (Dr. Van Ness testimony that his “understanding is citrate is chelating”); Ex. 1002 ¶ 23 (Dr. Edwards testimony that “citrate can chelate calcium ions so as to prevent coagulation”). The citric acid maintains the pH, counteracting the pH increase caused by the addition of sodium citrate. Ex. 2001 ¶ 169. The dextrose in ACD is “to counterbalance the sodium citrate in order to maintain the isotonicity of the sample environment.” *Id.* ¶ 170; Ex. 1054, 125:3–8 (Dr. Van Ness testimony that “dextrose may provide some isotonicity”). In short, ACD includes three compounds in aqueous solution—a chelator (sodium citrate), a pH buffer (citric acid), and dextrose (i.e., a type of glucose).

The ’720 expressly identifies glucose as an agent that impedes cell lysis. *See, e.g.*, Ex. 1001, 6:49-56 (“An agent that stabilizes cell membranes may be added to the maternal blood sample to reduce maternal cell lysis[,] including . . . glucose”); *see also, id.* at 11:57-63; 33:12-20; 214: 4-11;

219:55-62; 254: 1-8; 260:29-37; 264:14-21; 269:26-34; 274:36-43; 279:10-17; 285:1-8; and 290:44-51. This presents the question of whether glucose, which is expressly identified in the '720 patent as an agent that impedes cell lysis, is nonetheless precluded from being the claimed “agent” because it was introduced into the blood sample in a solution that also comprises sodium citrate and citric acid.¹⁸ We find that glucose is not so excluded.

The claims do not place any limitation on how the “agent” comes to be in the sample. They simply require that the “sample comprises” an “agent that impedes cell lysis, if cells are present.” It would seem incongruous that a blood sample containing a substance expressly called out as an “agent” in the specification of the '720 patent would be excluded from meeting the “agent” limitation simply because that substance was introduced to the sample as one component of a solution also comprising an anticoagulant/chelator. This is particularly true where there is no indication that the substance in question, in this case dextrose, plays any specific role in anticoagulation or chelation. *See, e.g.*, Ex. 1054, 125:11–12 (testimony of Dr. Van Ness on dextrose not being an anticoagulant).

In order for Patent Owner’s disavowal to extend to an individual component of an anticoagulant/chelator solution that itself plays no role in

¹⁸ As discussed *supra* § II.C, although the claims preclude an anticoagulant/chelator from being the claimed “agent,” they do not exclude an anticoagulant/chelator from being present in the sample. Indeed, as also discussed *supra* § II(C), the '720 discloses a method of processing maternal plasma in tubes that include *both* EDTA (an anticoagulant/chelator) and formaldehyde/formalin as a representative example of the invention.

anticoagulation or chelation, Patent Owner’s disavowal needed to make that unambiguously clear. *Poly-America, L.P.*, 839 F.3d at 1136 (“[T]he standard for disavowal is exacting, requiring clear and unequivocal evidence that the claimed invention includes or does not include a particular feature. Ambiguous language cannot support disavowal.”) (internal citation omitted).¹⁹

For the reasons discussed *supra* § II.C, it is clear that Patent Owner excluded anticoagulant/chelators from being the claimed “agent.” However, it is not clear that that exclusion extends to *all* of the individual components of anticoagulant solution—at least one of which is specifically identified in the ’720 patent as being an agent that impedes cell lysis. As discussed *supra* § II.C, during prosecution the Examiner considered two prior art references, Holodniy and Adams, each of which collected blood in tubes containing ACD, and found that they lacked a teaching of the claimed agent. However, there is no indication that in doing so, the Examiner considered the argument made here by Petitioner, i.e., that the dextrose component of ACD meets the “agent” limitation. In sum, Patent Owner’s disavowal does not

¹⁹ The district court in *Ravgen v. Laboratory Corporation of America Holding*, 6:20-cv-00969 determined that Lee did not disclose an agent that impedes lysis, if cells are present. Ex. 2306, 1. The district court granted summary judgment of no anticipation on this basis. *Id.* Although we are aware of the district court’s decision, it plays no role in our analysis because the standards of proof are different, the records are different, and, unlike proceedings in district court, there is no presumption of validity in a proceeding before the Board. Moreover, district court decisions are not binding on the Board. And, because it does not provide analysis explaining its reasoning, the district court’s decision is of limited persuasive value. *Id.*

unambiguously extend to the situation presented here. Even short of disavowal, we are also not persuaded that the ordinarily skilled artisan would interpret dextrose/glucose as excluded from possibly satisfying the “agent” limitation when other parts of the intrinsic evidence unambiguously list it as a membrane stabilizer and lysis inhibitor.

2. Has Petitioner established that the dextrose component of Lee’s ACD impedes cell lysis

By its plain language, claim 1 requires that “if cells are present,” as they are in Lee’s samples, the claimed agent “impedes cell lysis.” Thus, Petitioner cannot carry its burden simply by showing that the glucose (in its dextrorotary form, dextrose) is present in Lee’s blood samples. Rather, Petitioner must show that the dextrose in Lee’s samples “impedes cell lysis” “if cells are present.”

The current record includes voluminous evidence relating to whether Lee’s ACD impedes cell lysis. For readability, and to facilitate our analysis, we divide this evidence into three broad categories: 1) evidence relating to the capacity of glucose to act as a lysis inhibitor,²⁰ 2) evidence relating to differences between EDTA and ACD, and 3) evidence relating to cell-free DNA released in Lee’s serum samples as compared to cell-free DNA released in plasma samples.

²⁰ Evidence concerning the capacity of glucose to inhibit lysis is relevant to our analysis because it informs our inquiry into whether the dextrose in Lee impedes cell lysis. As discussed above, that Petitioner must do more than point to the capacity of dextrose in some circumstances and under certain conditions to impede cell lysis to prevail here.

a) Evidence relating to the capacity of glucose to act as a lysis inhibitor

As support that dextrose impedes cell lysis, Petitioner points to the multiple statements in the '720 patent describing glucose as a membrane stabilizer that can be used to inhibit cell lysis. Pet. 25 (citing Ex. 1001, 6:49-56; 11:57-63; 33:12-20; 214:4-11; 219:55-62; 254:1-8; 260:29-37; 264:14-21; 269:26-34; 274:36-43; 279:10-17; 285:1-8; and 290:44-51). We agree that these statements identify glucose as an agent that impedes cell lysis. *See e.g.*, Ex. 1001, 6:49–56 (disclosing that “[a]n agent that stabilizes cell membranes may be added to the maternal blood sample to reduce maternal cell lysis[,] including but not limited to . . . glucose”). These statements thus provide evidence that glucose (and thus dextrose) has the capacity to impede cell lysis.

Petitioner also points to evidence explaining how dextrose may function to prevent cell lysis, “by: (1) interacting with and stabilizing the cell membrane; (2) maintaining osmolarity so that the cells do not rupture; and (3) providing a nutrient source.” Reply 3 (citing Ex. 1002 ¶¶ 22–23; 2001 ¶ 169; Ex. 1059 ¶ 5; Ex. 1054, 125:3–127:12; 1057, 3–4, and Ex. 1056, 7–13, 15–19, 24). Patent Owner argues that the evidence as to how dextrose functions to prevent cell lysis should have been presented in the Petition. Sur-reply 16. On this record, we do not agree. While this evidence could have been included in the Petition, it is responsive to arguments raised in Patent Owner’s response. *See* PO Resp. 18–22. In any event, consideration of this evidence simply explains how and why dextrose has the capacity to impede cell lysis, and we have already found that the '720 patent provides

evidence that dextrose has this capacity. Under the circumstances, it was therefore reasonable for Petitioner, in the Petition, to rely on such disclosures of the patent itself. Further, consideration of how and why dextrose impedes cell lysis largely benefits Patent Owner, by providing reasons why dextrose might not inhibit cell lysis under the specific circumstance present in Lee.

Petitioner cites Dr. Edward's testimony to support that dextrose can interact with and stabilize the cell membrane. Reply 3 (citing Ex. 1002 ¶¶ 22–23). In the cited testimony, Dr. Edwards states, "it was known that carbohydrates have a role in membrane stabilization" and that dextrose "is a membrane stabilizer." Ex. 1002 ¶¶ 22–23. This is consistent with the teaching in the '720 patent that glucose is "[a]n agent that stabilizes cell membranes." *See e.g.*, Ex. 1001, 6:49–56. The evidence thus supports that dextrose can help to stabilize cell membranes. However, this evidence is tempered by evidence supporting that the "impact of carbohydrates on biological membranes is extremely complex" and that it is "extremely difficult to predict how solutes such as carbohydrates will affect biological membranes." Ex. 2231, 60:18–61:2, 83:14–18 (Dr. Edwards testimony). This is because "there's a finite but long – long list of variables that will impact the – the mechanism by which carbohydrate interacts with the cell membrane." *Id.* at 66:20–67:7; *see also id.* at 66:1–15 (identifying temperature, concentration and pressure as variables that play a role in how carbohydrates interact with biological membranes); *id.* at 68:1–17 (identifying prior and simultaneous exposure to other substances as variables that play a role in how carbohydrates interact with biological membranes).

Petitioner also provides evidence that dextrose helps prevent cell lysis by “maintaining osmolarity so that cells do not rupture.” Reply 3 (citing Ex. 2001 ¶ 169; Ex. 1056, 7–13, 24; Ex. 1059 ¶ 5). We agree that the cited evidence supports that dextrose has the ability to prevent cell rupture by maintaining isotonicity. In particular, Rapoport found that erythrocytes in an acid-citrate glucose solution underwent osmotic hemolysis at a slower rate than erythrocytes in a simple citrate solution. Ex. 1056, 613. However, Patent Owner introduces evidence, which we credit, that the failure to maintain isotonicity does not necessarily cause cell lysis. Ex. 2299, 237:4–16 (Dr. Edward’s testimony). For example, cells placed in a hypertonic solution may “shrink or kind of shrivel” rather than burst, and cells placed in a hypotonic solution may swell, but will not necessarily lyse. *Id.* at 245:3–16, 246:17–247:8.

Finally, Petitioner provides evidence that dextrose can help to prevent cell lysis by “providing a nutrient source.” Reply 3 (citing Ex. 1056, 15–19; Ex. 1057, 3–4; Ex. 1059 ¶ 5). We agree that the cited evidence supports that glucose can provide a nutrient source for cells. *See*, Ex. 1057, 16 (explaining that glucose “can support red cell metabolism during storage”); Ex. 1056, 604–605 (reporting “the glycolytic activity of erythrocytes ceased” earlier in simple citrate than ACD solution and hypothesizing that “in part the function of the glycolytic process is to yield the energy required for the maintenance of the cellular form”). However, the evidence does not persuasively support that dextrose serves as a nutrient source to impede cell lysis in Lee’s method. As Patent Owner points out, Lee stores its blood samples at 4°C, and Lee teaches that “WBCs [white blood cells] become

nonfunctional after storage at 4°C.” Ex. 1004, 277, 279. As a result, “any effect that glucose has by virtue of its metabolic capabilities to improve the health and stability of cells is not realized in any of the Lee experiments.” Ex. 1054:125:17–126:5 (deposition testimony of Dr. Van Ness); *see also*, Ex. 1059 ¶ 8 (Dr. Edwards testimony that “Lee does not disclose whether the blood cells stored in ACD were metabolizing glucose or not.”); Ex. 2231, 92:7–10 (Dr. Edwards testimony acknowledging that “when blood cells’ metabolism slows, [so] does their consumption of nutrients”).

b) Evidence relating to differences between EDTA and ACD

Patent Owner argues that a POSA would have understood, by comparing the performance of EDTA (an anticoagulant that lacks dextrose) with the performance of ACD (an anticoagulant with dextrose), that the dextrose in Lee’s samples did not act to impede cell lysis. PO Resp. 19–21. Patent Owner’s argument focuses on Lee’s “third experiment,” where “[p]lasma samples (ACD and EDTA) and serum samples from six normal donors were compared with respect to DNA yield after 0, 1, and 4 days of 4°C storage of blood.” Ex. 1004, 279. Lee reports that there was “no significant difference between EDTA and ACD plasma in concentration of cell-free genomic DNA on Day 0.” *Id.* at 280, Fig. 2A. Although Lee does not provide data or statements directly comparing EDTA and ACD for subsequent days, Lee states that “the concentration of cell-free genomic DNA in all six serum samples increased dramatically, whereas the increase in plasma DNA was comparatively much lower.” *Id.* at 279.

Based on our review of the evidence, we find that Lee provides only weak support for Patent Owner's argument that ACD and EDTA perform the same. Lee supports that at Day 0, there was no significant difference between ACD and EDTA. *Id.* at 280, Fig. 2A. However, Day 0 samples were prepared within 2 hours of phlebotomy. *Id.* at 277. The evidence supports that if samples are processed immediately, contamination from cell lysis may not be an issue. Ex. 1054, 49:11–50 (Dr. Van Ness testimony that background DNA was not recognized as a problem when samples were processed immediately, but that “as samples sat” they became “more and more contaminated . . . DNA was entering the sample for analysis.”); Ex. 1059 ¶ 6 (Dr. Edwards testimony that “one would not expect significant differences between EDTA, ACD and/or other membrane stabilizers in plasma prepared within 2 hours.”). The evidence of record thus supports that the data at Day 0 was taken too close to the phlebotomy to allow us to draw meaningful conclusions about the relative performance of EDTA and ACD with respect to lysis, much less evidence revealing what, if any, differential effect dextrose provided in this context.

Beyond Day 0, Lee teaches that cell-free DNA in serum samples increased “dramatically” while the increase in plasma DNA was “much lower.” Ex. 1004, 279. This supports that its serum samples released more cell-free DNA than plasma samples preserved with each of ACD and EDTA. Absent supporting data or a statement comparing ACD to EDTA, it supports only a weak inference that the performance of ACD was similar to that of EDTA. Here again, the evidence is lacking to draw a meaningful distinction between whatever anticoagulant/chelator effect EDTA and ACD may

provide versus what potential effect dextrose may have provided to impeding cell lysis.

Patent Owner directs us to studies that it characterizes as showing “no statistically significant difference between samples using ACD as an anticoagulant (with dextrose) and EDTA as an anticoagulant (without dextrose).” PO Resp. 20 n.5. For example, Patent Owner cites Nolte , which evaluated “the second versions of the manual AMPLICOR and the semiautomated COBAS AMPLICOR tests for hepatitis C virus” and found “no notable differences in clinical performance . . . between EDTA-plasma versus ACD-plasma” for both manual AMPLICOR and semiautomated COBAS AMPLICOR tests (Exs. 2236, Abstract, 4007). Patent Owner also cites Sebire, which investigated “the effects of conditions often encountered during handling, transit, and storage of blood specimens on the quantity of detectable human immunodeficiency virus (HIV) RNA in plasma” (Ex. 2237, Abstract). Sebire found “no statistically significant difference between $\log_{10}$ copy numbers in samples collected in ACD tubes and those of samples collected in EDTA tubes.” Ex. 2237, 494.

Sebire and Nolte are counterbalanced by two reference cited by Petitioner: Carter and Narayanan. Reply 4. Carter studied “the effects of anticoagulants [EDTA, ACD, and heparin] and cell preparation methods on lymphocyte forward-angle scatter (FSC), autofluorescence, and immunofluorescent staining.” Ex. 1060, Abstract. Carter found that “on day 1, . . . there was a significant increase in the number of granulocytes positive for PI (dead) in the EDTA anticoagulated blood. This cell death was less evident but present nonetheless with ACD and it was negligible with

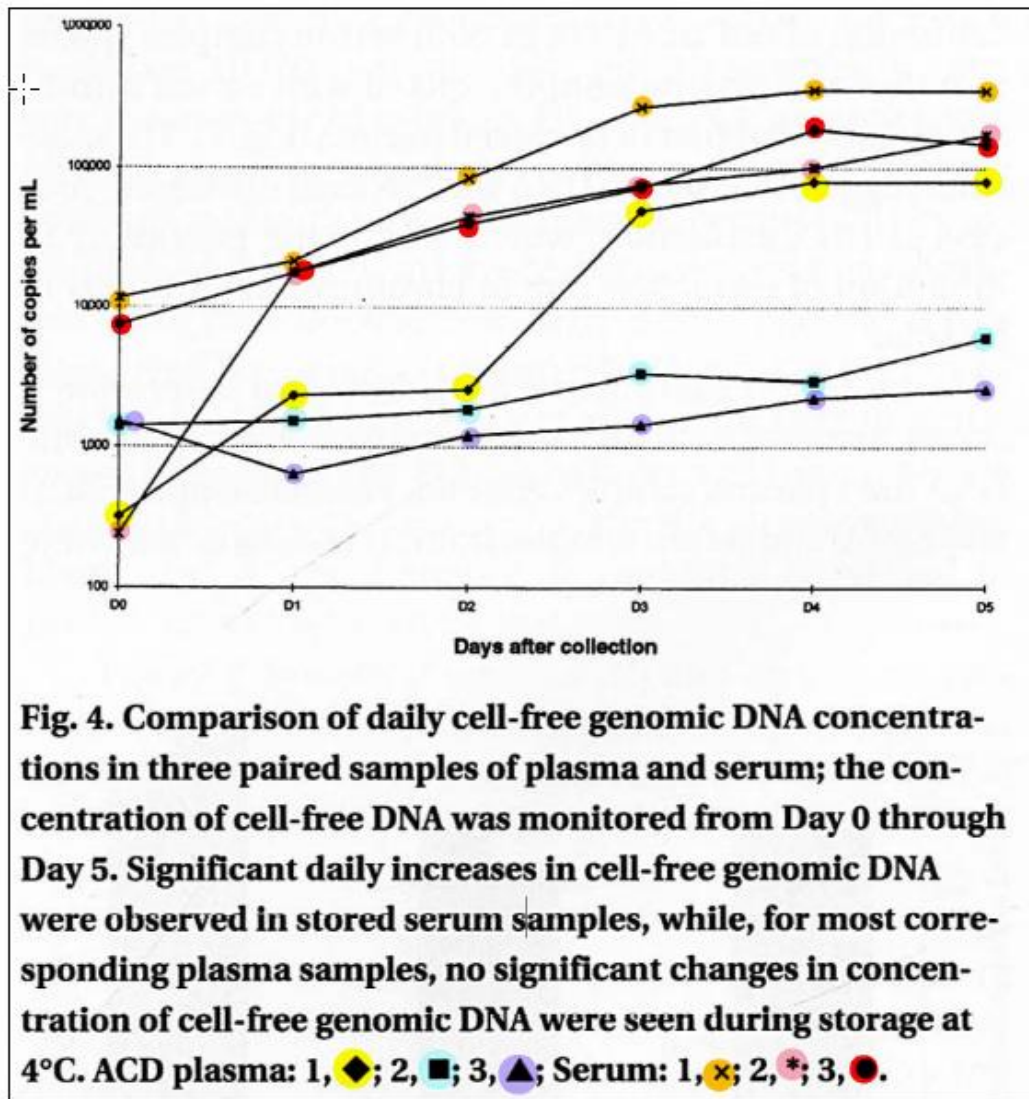
heparin.” *Id.* at 71. Narayanan is an article discussing preanalytic effects on laboratory results. Ex. 1058. In Narayanan’s discussion of anticoagulants used for blood collection, Narayanan reports that “[f]or flow cytometric analysis, ACD and heparin are superior to EDTA for maintaining viable WBCs overnight.” *Id.* at 446 (noting also that “if analysis is to be performed the same day, EDTA, ACD, and heparin give equivalent results”).

Although Sebire and Nolte provide some support for Patent Owner’s position by teaching that there is no difference between EDTA and ACD for purposes of detecting viral nucleic acid, Carter and Narayanan address to the effect of EDTA and ACD on cell viability, and thus speak more directly to the issue of whether ACD would be expected to impede cell lysis. On this basis, we give slightly more weight to Carter and Narayan than Sebire and Nolte.

Considering all of the evidence regarding the differences between EDTA and ACD together, we find that it weighs neutral in our analysis. That is, it neither supports nor diminishes the proposition that the dextrose in Lee impedes cell lysis.

c) *Evidence relating to cell-free DNA released in Lee's serum samples as compared to cell-free DNA released in plasma samples*

Petitioner contends that Figure 4 of Lee shows that dextrose must have some function in preventing cells from lysing. Reply 4. Figure 4, as annotated by Patent Owner, is reproduced below.



Ex. 1004, 280. Figure 4 is a graph plotting the daily cell-free DNA concentration of three samples of plasma stored with ACD and three

samples of serum stored without ACD. *Id.* The x-axis of the graph is “days after collection” and is scaled from 0 to 5. *Id.* The y-axis of the graph is “number of copies per ml” and ranges from 1,000 to 1,000,000 on a logarithmic scale. *Id.* The legend for the scale states that “significant daily increases in cell-free genomic DNA were observed in stored serum samples, while, for most corresponding plasma samples, no significant changes in concentration of cell-free genomic DNA were seen during storage at 4°C.” *Id.* Patent Owner has annotated the graph by color-coding the three sets of plasma data points yellow, blue, and purple and color-coding the three sets of serum data points orange, pink, and red. Sur-reply 13.

With reference to Figure 4, Petitioner points to the fact that cell-free DNA concentrations continued to increase in serum samples but not in plasma samples, which Petitioner urges is evidence that dextrose must have some preservative effect. Reply 4–5. The logic behind Petitioner’s argument is as follows. It was known that the process of clotting lyses blood cells, releasing DNA. See Ex. 1014, 1577 (“the amount of maternal circulatory DNA was significantly increased in serum samples compared with plasma because of the liberation of maternal DNA from maternal cells lysed during clotting”). By definition, serum is the fraction of blood that exists after coagulation is complete. Ex. 1059 ¶ 7 (citing Ex. 1055, 494). Thus, according to Petitioner, the cells present in the serum would “not be expected to undergo further coagulation-induced cell lysis.” Reply 4. (citing Ex. 1059 ¶ 7 (Dr. Edwards testimony to the same effect)). Nonetheless, cell-free DNA in the serum samples in Figure 4 of Lee continued to increase (Ex. 1004, 280), suggesting that the cells released cell-free DNA through lysis

caused by something other than coagulation (Ex. 1059 ¶ 7). In contrast, the ACD plasma samples, which, unlike the serum, were stored with ACD, did not show increased quantities of cell-free DNA. Ex. 1004, 280. According to Petitioner and Dr. Edwards, this suggests that the dextrose component of ACD played some role in preventing cells in the plasma samples from lysing. Reply 4–5; Ex. 1059 ¶ 7 (Edwards’ supplemental declaration, stating: “Given the known anticoagulant properties of citrate, and that citric acid functions as a buffer, a [POSA] would have understood that the dextrose components of ACD must function apart from anticoagulation.”).

Patent Owner presents two arguments as to why Figure 4 of Lee does not reflect that dextrose impeded cell lysis. Sur-reply 13–14. First, Patent Owner argues that Dr. Edwards conceded that some of the lysis in the serum samples “could be due to the fact that they had been previously coagulated” and Dr. Edwards could not “ascertain how much, *if any*, of the difference between serum and plasma DNA was attributable to the dextrose component of ACD.” *Id.* at 13–14 (emphasis added by Patent Owner). Second, Patent Owner argues that “Petitioner’s arguments are based on an inaccurate summary of the results shown in Figure 4.” *Id.* at 14. In particular, according to Patent Owner “ACD plasma sample #1 (yellow) released nearly the same amount of DNA as serum sample #2 (pink) across the six days.” *Id.* at 15.

On this record, we agree with Patent Owner that the strength of Petitioner’s evidence is diminished by the evidence that an unknown amount of the increase in cell-free DNA in serum samples could be attributable to prior coagulation. Dr. Edwards asserts that “the cells present in serum

would not be expected to be undergoing further coagulation so as to promote additional coagulation-induced cell lysis”²¹ (Ex. 1059, ¶ 7), but even though the cells are no longer undergoing coagulation, Dr. Edwards himself admits that some of the lysis in the serum samples could be attributed to the fact that they had previously coagulated. Ex. 2299, 217:22–218:9 (Dr. Edwards testimony that “[l]ysis that occurs in the serum samples – some of it could be due to the fact that they had been previously coagulated. I agree with that. . . . I don’t know how much.”). Accordingly, the evidence of record supports that some unknown portion of the increase in cell-free DNA observed in the serum samples of Lee may be attributable to prior coagulation.

We do not understand Dr. Edwards to have conceded to the possibility that dextrose played no role in the impeding cell-lysis in Lee’s plasma samples. Patent Owner cites the following testimony (Sur-reply 13), emphasizing the words “if any” in counsel’s question to Dr. Edwards:

Q: And Lee does not provide any way of knowing what amount, *if any*, of diminished increase in cell-free DNA is due to the presence of dextrose, right?

A: No. I can’t separate the amounts in Lee’s calculations because there is not any information to do that. I just have to look at the data and realize that, as I stated in the supplemental declaration, that these are in log scale. So it’s a huge change in those samples [i.e., the serum samples], and there is some increase, small, in the ACD [plasma] tubes, which is perfectly consistent with what one would expect when you store plasma

²¹ Patent Owner argues that this statement lacks evidentiary support (Sur-reply 11), but Dr. Edwards testimony is consistent with the definition of serum (Ex. 1005, 494 (“Serum is the fluid left after plasma has clotted, and so it can no longer clot.”)).

in an ACD tube, when you include the dextrose or because you have included the dextrose.

Ex. 2299, 217:4–20 (objection omitted) (emphasis added to mirror emphasis in Patent Owner’s briefing). After answering counsel’s question “no,” Dr. Edwards stated that he could not parse out relative contribution of dextrose. He then explained that the “huge” increase in cell-free levels in serum and the smaller increase in plasma was consistent with what one would expected from the presence of dextrose. Considered in context, it appears Dr. Edwards was admitting only that he could not quantify the amount of dextrose’s contribution. He did not necessarily admit, in a more complete context, that dextrose could have played no role.

Our interpretation of this testimony is consistent with Dr. Edwards’ testimony elsewhere in his deposition where he described coagulation as the minor variable and dextrose as the major variable influencing the level of cell-free DNA:

We have got a couple of variables here. The ACD plasma tubes have the acid citrate – citric acid and citrate functioning to prevent coagulation, and then you have the dextrose. And in the serum tubes, you will not have the anticoagulant, and you will not have the dextrose. And so the argument I am setting out in the supplementary declaration is that, yes, there are multiple variables at play.

However, since they have been coagulated already, we will say that that’s going to be the minor variable. And the major variable will be the dextrose, the membrane stabilizer, because that had been known for nearly 100 years, as of today, that is, 80 years, I guess, as of the time of the ’720 patent to be an additive that would stabilize and preserve cells.

Ex. 2299, 212:20–213:25. Thus we understand Dr. Edwards to opine that most of the difference between cell-free DNA levels in serum and plasma samples is attributable primarily to dextrose. However, we struggle to reconcile this testimony with Lee’s teaching that “[t]he most likely explanation for the higher levels of [cell-free] genomic DNA in serum than in plasma is that the process of clotting lyses WBCs [white blood cells].” Ex. 1004, 279; *see also id.* at 281 (“most cell-free genomic DNA in serum samples is generated during the process of clotting”). Nowhere does Lee hint at the significance of dextrose in the samples or attribute any difference in the observed differences in DNA levels to any specific cause beyond clotting.

We turn now to Patent Owner’s second argument, i.e., that “plasma sample #1 (yellow) released nearly the same amount of DNA as serum sample #2 (pink).” Sur-reply 14. In considering the strength of this argument, we rely on Lee’s characterization of the increases in cell-free DNA. Specifically, Lee characterized the increases in cell-free genomic DNA in its serum samples as “[s]ignificant” and characterized “most corresponding plasma samples” as showing “no significant changes.” Ex. 1004, 280. Particularly when compared to its unqualified characterization of serum samples, Lee’s statement that “most” of the plasma samples showed “no significant changes” suggests that one of the three plasma samples did, in fact, show significant changes. That one of the three plasma samples released significant quantities of cell-free genomic DNA diminishes the strength of Dr. Edwards’ testimony that some portion of the difference in cell-free DNA levels is directly attributable to dextrose.

d) *Conclusion*

In considering whether Petitioner has demonstrated by a preponderance of the evidence that the dextrose in Lee impedes cell lysis, we are mindful that the challenged claims do not require any particular degree of lysis impedance. Thus, the dextrose in Lee would be considered to be the “claimed agent” if it even minimally impeded cell lysis. *Id.* With this in mind, we consider the totality of the evidence discussed *supra* §§ II.E.2.a–c. We find that Petitioner has not demonstrated by a preponderance of the evidence that the dextrose in Lee’s blood samples impedes lysis. Our reasoning is as follows.

Petitioner’s evidence that the dextrose in Lee impedes cell lysis rests primarily on the testimony of Dr. Edwards that the difference between cell-free DNA released between Lee’s serum samples and its plasma samples is attributable to dextrose. Ex. 1059 ¶ 7. This testimony is supported by the evidence provided in the ’720 patent that glucose is listed as a membrane stabilizer and lysis inhibitor. *See e.g.*, Ex. 1001, 6:49-56. However, as discussed *supra* § II.E.2.a, glucose’s potential to impede cell lysis is circumstance specific. Here, Lee’s samples were stored at 4°C, marginally above freezing, rendering Lee’s cells “nonfunctional.” Ex. 1004, 277, 279. As discussed *supra* § II.E.2.a, this supports that the dextrose in Lee would not prevent lysis by providing nutrients to cells. The evidence also calls into question whether dextrose, as used in Lee, would prevent lysis by stabilizing the membrane of cells in Lee’s samples. *See supra* § II.E.2.a (discussing how the interaction between membranes and carbohydrates is complex and dependent on a number of variables, including temperature). Petitioner and

Dr. Edwards do not persuasively address temperature, isotonicity, or any other variable that might affect the ability of dextrose to impede cell lysis under the conditions present in Lee.

When pressed on cross-examination, Dr. Edwards testified that the difference in cell-free DNA levels seen in serum as compared to plasma is attributable primarily to dextrose in the plasma samples. Ex. 2299, 212:20–213:25. But this testimony is called into question by the fact that one of Lee’s plasma samples (which was stored with dextrose) experienced a significant increase in cell-free genomic DNA. Ex. 1004, 280 (discussed *supra* § II.E.2.c). It is further called into question by Lee’s teaching that most of the difference between cell-free genomic DNA observed in serum and plasma is attributable to prior coagulation. Ex. 1004, 279, 281. Dr. Edwards also recognized that prior coagulation was responsible for some portion of the difference in cell-free genomic DNA observed, but was unable to determine how much was attributable to dextrose and how much was attributable to prior coagulation. Ex. 2299, 217:4–20. This raises the question: when the contribution attributable to coagulation is removed, is whatever remains of the observed increase in cell-free genomic DNA in serum even significant, or was it, like two of the three plasma samples insignificant? Absent persuasive testimony parsing out relative contributions, we cannot say on this record. Under these circumstances, Petitioner has not persuaded us that there was a difference between cell-free DNA released by serum as compared to plasma that was attributable to dextrose. We find the preponderance of the evidence does not support that the dextrose in Lee impeded cell lysis.

3. Conclusion on Ground 1

Petitioner has not established by a preponderance of the evidence that Lee discloses an “agent that impedes cell lysis, if cells are present” as required by claim 1 and the claims depending therefrom. Accordingly, Petitioner has not established by a preponderance of the evidence that claims 1–6, 21, and 23 are anticipated by Lee.

F. Obviousness over Lee

Petitioner asserts that claim 22 would have been obvious over Lee. Pet. 31–33. As with Ground 1, Petitioner relies on the dextrose in Lee’s ACD as disclosing an “agent that impedes cell lysis, if cells are present.” Pet. 33 (referring back to Petitioner’s proof for Ground 1). For the reasons discussed with respect to Ground 1, Petitioner has not established by a preponderance of the evidence that Lee discloses “an agent that impedes cell lysis, if cells are present.” Accordingly, Petitioner has not established by a preponderance of the evidence that claim 22 would have been obvious over Lee.

G. Obviousness over Lee and Sorenson

Petitioner asserts that claim 7–9 and 24–30 would have been obvious over the combination of Lee and Sorenson. Pet. 33–44. As with Ground 1, Petitioner relies on the dextrose in Lee’s ACD as disclosing an “agent that impedes cell lysis, if cells are present.” Pet. 36 (referring back to Petitioner’s proof for Ground 1). For the reasons discussed with respect to Ground 1, Petitioner has not established by a preponderance of the evidence that Lee discloses “an agent that impedes cell lysis, if cells are present.”

Accordingly, Petitioner has not established by a preponderance of the evidence that claims 7–9 and 24–30 would have been obvious over the combination of Lee and Sorenson.

H. Obviousness over Sorenson and Rao

Petitioner asserts that claim 1–9 and 21–30 would have been obvious over the combination of Sorenson and Rao. Pet. 44–63. As we discuss in our analysis below, this ground turns on whether Petitioner has met its burden to establish, by a preponderance of the evidence, that the POSA would have found it obvious to combine and modify the asserted cited art. More specifically, it turns on whether it would have been obvious to modify Sorenson’s method of detecting specific sequences of the cystic fibrosis and *K-ras* genes in plasma DNA to include formaldehyde as a fixative agent, as allegedly disclosed in Rao, in the manner proposed by Petitioner.²²

1. Petitioner’s position

As discussed *supra* § II.D.2, Sorenson discloses detection of specific gene sequences in plasma or serum. Petitioner acknowledges, however, that Sorenson “does not specifically describe the use of an agent that inhibits cell lysis.” Pet. 48. Petitioner contends that the combination of Sorenson and

²² In its Response, Patent Owner argued that Rao does not qualify as prior art under 35 U.S.C. §§ 102(a) or 102(e). PO Resp. 25–37. We need not decide whether Rao qualifies as prior art to decide this case because counsel for Patent Owner withdrew the argument that Rao did not qualify as prior art at oral argument (Tr. 58) and because, even if Rao is prior art for all claims, Petitioner’s challenge based on Sorenson and Rao fails for other reasons, as discussed herein.

Rao addresses this deficiency. *Id.* at 45–49. Specifically, Petitioner argues that Sorenson discloses that “serum may contain artificially high levels of soluble (cell-free) DNA caused by the release of cellular DNA during clotting of whole blood, thereby causing assays of cell-free DNA in serum to yield falsely high results” while Rao discloses “the use of formaldehyde and formaldehyde donors such as imidazolidinyl urea as cell lysis inhibitors and/or membrane stabilizers to preserve circulating tumor cells in blood specimens.” Pet. 48. According to Petitioner, Rao shows that “the addition of . . . stabilizers minimizes cell damage and preserves sample quality.” *Id.* at 49.

Based on these disclosures, Petitioner argues that a POSA would have been motivated to use formaldehyde or a formaldehyde donor in Sorenson’s detection method, explaining:

A [POSA] would have been motivated to try various agents, such as the agents (e.g., formaldehyde or formaldehyde donors) described in Rao, as an alternative to or in combination with EDTA, to inhibit the lysis of cells in a blood sample undergoing analysis for cell free nucleic acids to decrease assay background for improved quantitation of free nucleic acid as disclosed by Sorenson.

Id. at 50. According to Petitioner, Rao shows “a reduction in cell lysis for plasma isolated from tubes containing formaldehyde.” *Id.* Thus, Petitioner argues, a POSA would reasonably have expected that the “use of formaldehyde, in place of or in addition to the EDTA used in Sorenson’s methods would have been successful in stabilizing cells and inhibiting cell lysis in the sample, thereby facilitating detection of free nucleic acids.” *Id.* at 50–51 (citing Ex. 1002 ¶ 86 (Dr. Edwards testifying to similar effect)).

2. Patent Owner's position

Patent Owner argues that a POSA would not have been motivated to use Rao's formaldehyde or a formaldehyde donor in Sorenson's method, and would not have reasonably expected success in doing so, because it was known that formaldehyde can have detrimental effects on nucleic acid products. PO Resp. 47–52. According to Patent Owner, multiple sources, including Rao, Srinivasan, and Swenberg,²³ reported on formaldehyde's detrimental effect on nucleic acids, undermining Petitioner's reasons for adding it to Sorenson's method and diminishing the expectation that adding formaldehyde would be successful. *Id.* at 48–50; Ex. 2078 ¶¶ 152–158.

Patent Owner explains that prior to the claimed method, formaldehyde had been used as “a fixative for applications studying *cells*, not non-cellular material such as cell-free DNA,” which many researchers, including Rao, had until then discarded. *Id.* at 50 (citing Ex. 2078 ¶¶ 155–156). Thus, Patent Owner asserts, “researchers had not concentrated on how any compounds, and certainly not formaldehyde, would affect downstream cell-free DNA analysis, including isolating cell-free DNA from plasma, amplifying it with PCR, and detecting the amplified cell-free DNA.” *Id.* Patent Owner concludes that “Petitioner's proposed combination is nothing

²³ Patent Owner cites several other references as evidencing the problems with formaldehyde's use, including damage to nucleic acids. Although such references do appear to report consistently on drawbacks with formaldehyde, many of those references post-date the filing date of the '720 patent by years. *See, e.g.*, PO Resp. 50–52 (citing Ex. 2155 (published in 2005); Ex. 2047 (published in 2018)). For purposes of this decision, we do not rely on these references.

more than a hindsight reconstruction of the claims,” and a POSA “would have feared that adding Rao’s stabilizers to Sorenson’s methods could compromise the small amounts of cancerous cell-free DNA” present in Sorenson’s samples. Sur-reply at 21.

3. Analysis

On this record, we agree with Patent Owner that Petitioner has not carried its burden to establish that a POSA would have been motivated to add formaldehyde to Sorenson’s method of detecting cell-free DNA in plasma.

A key question presented in this case is whether a POSA would have been concerned with formaldehyde’s potential effects on DNA, and cell-free fetal DNA in particular. Cellular DNA and cell-free DNA are not the same thing. It is not simply the location of the DNA as intracellular versus extracellular; to the contrary, there are structural and biochemical differences. For example, cell-free DNA is typically short fragments compared to cellular DNA which is typically long, genomic DNA. Ex. 2299, 26:11–14 (identifying the size as a difference between cell-free and cellular DNA and explaining that cellular DNA is “thought to be complete in terms of entire chromosomes” and that even when it is released “it would still be large fragments”); *id.* 34:15–17 (“[c]ell free DNA is going to be much more resistant to mechanical damage because it’s already pretty short”). There are also differences in whether the DNA is wrapped around histone-forming chromatin. *Id.* at 25:22–26:9 (identifying “how it’s wrapped around histone-forming chromatin” as a “difference[] in the physical state of cell-free versus cellular DNA” and confirming that “cell-

free DNA is not wrapped around histone-forming chromatin”); 28:18–23 (“cell-free DNA would basically be considered just free plain DNA, not wrapped around any types of chromatin structures”).²⁴

Recognizing these differences, we consider how the POSA would expect formaldehyde to affect cell-free DNA. Srinivasan provides strong evidence that formaldehyde was known to have detrimental effects on nucleic acids. It discloses that attempts to extract usable DNA from formaldehyde fixed tissues have only been “variably successful,” explaining that “considerable evidence suggests that formaldehyde induces DNA degradation” with “few” studies reporting yield of high-molecular weight DNA. Ex. 2150,²⁵ 1964 (also disclosing that “formaldehyde initiates DNA denaturation” and causes a high frequency of “sequence alteration”). In view of these concerns, Srinivasan suggests that one way to “overcome the problems of formaldehyde is to use an alternative fixative that is better suited for the preservation of nucleic acids.” *Id.* at 1966.

Rao also supports that formaldehyde was known to damage nucleic acids. Rao teaches that formaldehyde “has been reported to react with nucleic acid bases, particularly adenine, to reversibly form hydroxymethylol derivatives and methylene bridges thereby irreversibly crosslinking nucleic acids.” Ex. 1006, 4; Ex. 2078 ¶ 153. Similarly, Swenberg cites a report that “formaldehyde could introduce genetic alterations” and discusses studies

²⁴ Patent Owner submits additional evidence that cell-free DNA is different in that it is only available in limited quantities, can be diluted due to lysis, and is easily damaged. Ex. 2086 ¶ 4; *see also* Ex. 2299, 157:14–161:21.

²⁵ Srinivasan was also entered into the record as Exhibit 1065.

demonstrating that formaldehyde can induce “single-stranded DNA breaks,” “DNA-protein crosslinks,” “sister chromatid exchanges,” and “chromosome aberrations and mutations.” Ex. 2139, 945.

Considering the evidence of record, including, particularly, Srinivasan, Rao, and Swenberg, we agree with Patent Owner that the evidence on this record supports that formaldehyde was known to damage nucleic acids. Petitioner’s expert, Dr. Edwards, agrees. Ex. 2299, 92:11–17 (Dr. Edwards testimony agreeing that “there were concerns about formaldehyde damaging DNA”), 33:13–16 (Dr. Edwards testimony agreeing that “DNA inside of cells and DNA outside of cells are both susceptible to damage from formaldehyde”).

Having determined that formaldehyde was known to damage nucleic acids, we next consider how this impacts the motivation of the POSA to use formaldehyde in Sorenson’s methods. The Petition is silent about the potential effects of formaldehyde on cell-free DNA, focusing instead on the expectation that formaldehyde would inhibit cell lysis:

In view of Rao’s disclosure showing a reduction in cell lysis for plasma isolated from tubes containing formaldehyde, a [POSA] would have had a reasonable expectation that use of formaldehyde, in place of or in addition to the EDTA used in Sorenson’s methods would have been successful for stabilizing cells and inhibiting cell lysis in the sample, thereby facilitating detection of free nucleic acids in the sample.

Pet. 50 (citing Ex. 1002 ¶ 86) (internal citation omitted).

In its Reply, Petitioner responds to Patent Owner’s argument that a POSA would have had concerns about the use of formaldehyde by arguing that a POSA “would have appreciated that fixatives could be used and would

have known how to tailor their use to avoid DNA damage.” Reply 20. As support, Petitioner cites the testimony of Dr. Edwards, who testifies:

Srinivasan . . . summarized how formaldehyde can be used without damage to DNA, including “a prefixation time lag of less than two hours, use of cold 10% neutral formalin, low salt concentration, a duration of fixation of 3-6 hours, EDTA additive at a concentration of 20-50 mmol/L, and avoiding a low pH.”

Ex. 1059 ¶ 25. Petitioner’s proposal for mitigating potential damage from formaldehyde, as explained by Dr. Edwards, thus involves, *inter alia*, limiting exposure of nucleic acid to formaldehyde to a period of 3–6 hours.

This is problematic for Petitioner at least because the proposed combination of Sorenson and Rao requires that the cell free DNA be exposed to formaldehyde for much longer than 3–6 hours. In his deposition, Dr. Edwards explained that in the proposed Sorenson/Rao combination, DNA would be exposed to formaldehyde for 18–24 hours:

Q. So what I intended to ask was the total duration of exposure to a stabilizer in combining Sorenson and Rao in your combination would be potentially upwards of 24 hours, right?

A. I think, yes. You are referring to the 24 hours because of the dialysis step. Is that where you are getting the 24 hours?

Q. That seems to be one aspect of the steps that Sorenson is laying out here so . . .

A. So, yes. If you practice exactly as described in Sorenson, you would be doing this dialysis, and I think they did dialyse[] it against three changes of this ten millimolar Tris EDTA or TA buffer at 4 degrees. So 18 to 24 hours with the three changes at that time period. So, yes. You would have – the way the combination would work is, yes. It would be mixed together with the stabilizer through the previous steps starting prior to the centrifugation.

Ex. 2299, 104:11–105:9; Sur-reply 21–22 (citing this testimony).

Accordingly, at least one of the allegedly ameliorative steps proposed in Petitioner’s Reply is incompatible with Petitioner’s proposed combination. For at least this reason, we do not credit Dr. Edwards’ testimony, or Patent Owner’s argument, that the POSA would have known how to tailor the use of formaldehyde to address concerns regarding damage to cell-free DNA within the proposed combination of Sorenson and Rao.

Petitioner also responds to concerns about formaldehyde’s effect on Sorenson’s cell-free DNA by arguing that the use of formaldehyde was “fully appreciated.” Reply 20. As evidence, Petitioner cites Srinivasan, Rao, and U.S. Patent No. 5,849,517 (“the Streck patent”).

Petitioner’s citation to Srinivasan is to its recommendations on how to use formaldehyde as a “tissue nucleic acid fixative.” *Id.* We discussed this above in connection with Dr. Edwards’ testimony on “how formaldehyde can be used without damage to DNA” and will not repeat it here.

Petitioner cites Rao as disclosing “the use of CytoChex™ BCTs [blood collection tubes] for stabilization of normal WBCs [white blood cells] and CTCs [circulating tumor cells].” *Id.* While Petitioner is correct that Rao discloses the use of CytoChex™, it does not do so in the context presented here – i.e., for the preparation and analysis of cell-free nucleic acids. Rao was concerned with preserving rare blood *cells*, particularly circulating tumor cancer *cells*, and paid no heed to the effect of its preservatives on cellular DNA, much less cell-free DNA. Ex. 1006, 2 (“This invention relates generally to the field of cell and blood stabilization, more particularly to the stabilization of rare cells in blood specimens and most

particularly to stabilization of circulating tumor cells (CTC) in whole blood for subsequent enrichment and analysis.”); Ex. 2078 ¶¶ 146–147. Indeed, Rao discards the portion of the sample that might have contained any cell-free DNA. *Id.* at 13 (“After mixing, the samples were centrifuged at 800x g for 10 minutes to remove plasma.”); Ex. 2078 ¶ 146 (“Rao discards the plasma along with cell-free DNA prior to analysis”). Moreover, Rao teaches that “[f]ormaldehyde *released from these so-called formaldehyde donors* has been reported to react with nucleic acid bases . . . thereby irreversibly crosslinking nucleic acids.” Ex. 1006, 4; cited at PO Resp. 48. Patent Owner does not persuasively address this concern. Accordingly, we are not persuaded that the known use of CytoChex™ in Rao would have assuaged concerns a POSA would have had about the effect of formaldehyde on cell-free nucleic acid.

As for the Streck patent, Petitioner contends that it shows that “formaldehyde donors were already known to allow for DNA testing.” Reply 20. As support, Petitioner and its expert cite only to the title of the Streck patent, which describes “[p]reserving . . . [n]ucleic acids.” Reply 20; Ex. 1059 ¶ 25. Petitioner does not direct us to, and we do not find, evidence that the Streck patent considered the effect of formaldehyde donors on *cell-free* DNA. *See generally*, Ex. 1063 (“the Streck patent”); *see also*, Ex. 2086 ¶¶ 1, 6, 8 (Declaration of Streck’s Director of Research and Development, stating that the Streck patent was directed to “a product called Streck Cell Preserve” and that prior to 2010, there was “no literature to suggest use of Streck Cell Preserve or Cyto-Chex® BCT for use in [cell-free] fetal DNA recovery”). Particularly given the differences between cellular DNA and

cell-free DNA (*see supra*), we are not persuaded that the use of formaldehyde donors in the Streck patent would have assuaged concerns a POSA would have had about the effect of formaldehyde (whether added directly or via donors) on cell-free nucleic acid.

At oral argument, Petitioner argued that that concerns about formaldehyde would not apply to formaldehyde donors, including, specifically CytoChex. Tr. 76. When asked to identify that argument in its briefing, Petitioner directed us to page 20 of its Reply. *Id.* at 76, 88. We do not find this argument on page 20 of Petitioner’s Reply. Certainly, the Reply discusses Rao and the Streck patent, which discuss CytoChex and formaldehyde donors, but nowhere do we find argument seeking to draw a distinction between formaldehyde and formaldehyde donors as relates to their expected effects on cell-free nucleic acid. Accordingly, this argument is waived. Even if it were not waived, absent further explanation from Petitioner, the evidence cited on page 20 of Petitioner’s Reply, all of which is discussed above, does not persuasively support such a distinction. *See e.g.*, Ex. 1006, 4 (“Formaldehyde ***released from these so-called formaldehyde donors*** has been reported to react with nucleic acid bases . . . thereby irreversibly crosslinking nucleic acids.”) (emphasis added).²⁶

²⁶ The Petition asserts that “a [POSA] would have had a reasonable expectation that use of formaldehyde, in place of or in addition to the EDTA used in Sorenson’s methods, would have been successful for stabilizing cells and inhibiting cell lysis in the sample.” *Id.* The Petition does not assert that the POSA would have had an expectation of success for any agent other than formaldehyde. *Id.* The testimony of Petitioner’s expert, Dr. Edwards,

In sum, the evidence of record supports that formaldehyde was known to have detrimental effects on nucleic acids. The record does not support that a POSA would have known how to tailor the use of formaldehyde to ameliorate the expected detrimental effects of formaldehyde on nucleic acid in a manner consistent with Petitioner's proposed combination. The record also does not support that known uses for formaldehyde would have addressed concerns a POSA might otherwise have had regarding the use of formaldehyde for the preparation and analysis of cell-free nucleic acids. Considering all of the evidence of record relating to the effects of using formaldehyde in the method of Sorenson, we agree with, and credit, the testimony of Dr. Van Ness who explains that "concerns regarding formaldehyde's potential detrimental effects on DNA would have dissuaded a POSA from using formaldehyde" for the preparation and analysis of cell-free nucleic acids. Ex. 2078 ¶ 155.

4. Conclusion

Petitioner has not established by a preponderance of the evidence that the POSA would have been motivated to use formaldehyde in Sorenson's method for detection of specific gene sequences in plasma or serum. Petitioner has thus not met its burden to establish that the combination of

similarly does not posit an expectation of success for any agent other than formaldehyde. Ex. 1002 ¶ 86. To the extent Petitioner contends that a POSA would have different expectations as to how formaldehyde and a formaldehyde donor would affect nucleic acids, further explanation was required as to why a POSA would reasonably have expected success for each.

Sorenson and Rao renders obvious the use of an “agent that impedes cell lysis, if cells are present” as required by claim 1 and the claims depending therefrom. Accordingly, Petitioner has not established by a preponderance of the evidence that claims 1–9, and 21–30 would have been obvious over the combination of Sorenson and Rao.

III. MOTIONS FOR PROTECTIVE ORDER AND TO SEAL

Patent Owner moves for entry of a stipulated protective order and for an order sealing portions of the Patent Owner Response (Paper 21), all of Exhibits 2170–2173, and portions of Exhibit 2080. *See* Paper 21 (attaching stipulated protective order as Appendix A). That motion is unopposed.

A party may move to seal confidential information including, *inter alia*, sensitive commercial information. Consolidated Patent Office Trial Practice Guide, 19 (Nov. 2019); 37 C.F.R. § 42.54. It is the movant’s burden to show good cause for sealing such information, and we balance the party’s asserted need for confidentiality with the strong public interest in open proceedings. *Argentum Pharms. LLC v. Alcon Research, Ltd.*, IPR2017-01053, Paper 27 at 4 (PTAB Jan. 19, 2018) (informative).

Patent Owner provides a sufficient explanation for sealing the relevant portions of the Patent Owner Response and the identified exhibits. Exhibits 2170–2173 are license or settlement agreements between Patent Owner and third parties setting forth, for example, payment terms and sales data. Patent Owner states that it produced those agreements with the permission of third parties on the condition that the agreements remain sealed. Paper 21, 3–4. The subject portions of the Patent Owner Response and Exhibit 2080 include discussions about those agreements. Patent Owner

has also provided redacted versions of Exhibit 2080 and the Patent Owner Response (Paper 20) so the record may remain clear and reasonably open.

Patent Owner has established good cause for sealing Exhibits 2170–2173, portions of Ex. 2080, and the Patent Owner Response. The stipulated Protective Order includes minor changes from our default language. That Protective Order, Appendix A to Paper 21, is entered.

Petitioner also filed a Motion to Seal (Paper 28). Petitioner contends that portions of its Reply (Paper 29), and portions of Exhibit 1067 (deposition transcript of Paul K. Meyer) should be sealed because those papers include information that Patent Owner considers contain confidential business information (e.g., licensing practices). Paper 28, 1. Petitioner has also filed public redacted versions of its Reply (Paper 30) and Exhibit 1067. For the same reasons discussed above regarding Patent Owner’s Motion to Seal, Petitioner has shown good cause for granting its motion. Petitioner’s Reply (Paper 29) and Exhibit 1067 are sealed.

IV. CONCLUSION

In Summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
1–6, 21, 23	102(b)	Lee		1–6, 21, 23
22	103(a)	Lee		22
7–9, 24–30	103(a)	Lee, Sorenson		7–9, 24–30
1–9, 21–30	103(a)	Sorenson, Rao		1–9, 21–30
Overall Outcome				1–9, 21–30

V. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that the Petitioner has not proved by a preponderance of the evidence that claims 1–9 and 21–30 are unpatentable;

FURTHER ORDERED that Patent Owner’s Motion for Entry of Stipulated Protective Order (Appendix A to Paper 21) and to Seal (Paper 21) is *granted*;

FURTHER ORDERED that Petitioner’s Motion to Seal (Paper 28) is *granted*; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2021-01026
Patent 7,727,720 B2

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